

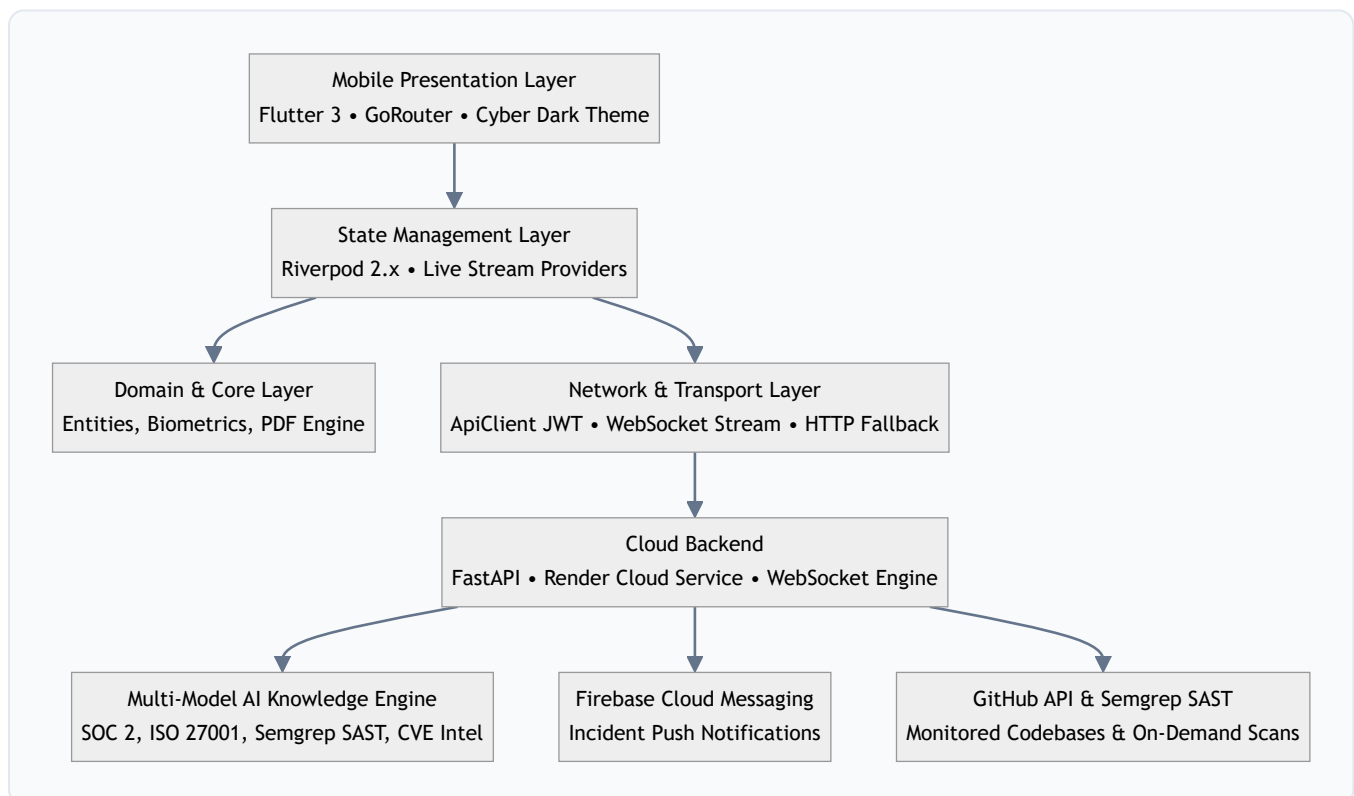
# SecurePulse

 FLUTTER 3.X DART 3.X FASTAPI CLOUD BACKEND FIREBASE FCM PUSH

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**SecurePulse** is a next-generation enterprise cybersecurity and SOC mobile operations platform built for security analysts, incident responders, and CISOs. Designed with a high-contrast dark cyberpunk aesthetic, hardware-backed biometric security, live WebSocket threat streaming, dynamic AI security assistance, real GitHub SAST repository scans, and one-tap executive PDF compliance report exports.

## Architecture & Tech Stack



### Core Technologies

- **Mobile Frontend:** Flutter 3 (Dart 3), Riverpod State Management, GoRouter, `flutter_animate`, `fl_chart`.

- **Cloud Backend:** Python 3.11, FastAPI, Uvicorn, WebSockets, JWT Authentication, Render Cloud.
- **Push Notification Pipeline:** Google Firebase Cloud Messaging (FCM) & `firebase_admin` SDK.
- **Document Engine:** Pure vector `pdf` engine with native `share_plus` and `path_provider` export.
- **Hardware Security:** `local_auth` biometric authentication (Fingerprint / Face ID).

## Key Platform Capabilities

### 1. Executive Security Operations Dashboard

- Real-time **Security Posture Score** (94/100, Grade **A+**) with animated gauge rings and vulnerability charts.
- Live system status monitors for Cloud API, WebSockets, GitHub integration, and AI engine.
- Instant access to quick actions: One-tap SAST scans, SOC incident triage, AI chat, and compliance reports.

### 2. Real-Time WebSocket SOC Threat Streaming

- Subscribes to live WebSocket telemetry ( `wss://.../ws/alerts` ) for instant incident detection.
- Resilient architecture with **automatic HTTP long-polling fallback** if network proxies block raw sockets.
- Incident severity badges (Critical, High, Medium, Low), attack vector classifications, and IP origin telemetry.

### 3. GitHub Repository Integration & Live SAST Scans

- Monitored codebases include `NATTOMR/secureguard-mobile` and `NATTOMR/secureguard-backend`.
- One-tap "**Trigger Immediate SAST Scan**" dispatches real-time Semgrep & Secret scanning across cloud repositories.
- Live telemetry sync updates repository health grades and findings counts on your phone.

### 4. Dynamic AI Security Copilot ( `SecurePulse AI Agent` )

- Expert cybersecurity knowledge engine for **SOC 2 Type II**, **ISO/IEC 27001:2022**, and zero-day threat patches.
- Generates tailored remediation patches, `iptables` / `ufw` / Cloudflare firewall rules, and custom Semgrep YAML rules.

### 5. Executive PDF Compliance Report Export

- One-tap generation of official audit PDFs for:
- **SOC 2 Type II** (Trust Services Criteria CC6.1 - CC8.1)
- **ISO/IEC 27001:2022** (Annex A.5 - A.8)
- **PCI-DSS v4.0 & HIPAA Security Rule**

- Includes cryptographic SHA256 audit verification stamps and direct export via the native Android Share Sheet (Email, Slack, WhatsApp, Google Drive).

## 6. 🎨 Custom Cyber Shield Launcher & Splash Branding

- Custom dark obsidian metallic app icon with a **neon cyan holographic cyber defense shield and circuit lock core**.
- Android adaptive launcher icons ( `mdpi` , `hdpi` , `xhdpi` , `xxhdpi` , `xxxhdpi` ) and shimmering animated splash screen.

## 📁 Project Directory Structure

```
secureguard-mobile/
├── android/                # Android native configuration & Gradle scripts
│   ├── app/
│   │   ├── google-services.json # Firebase configuration
│   │   └── src/main/res/        # Custom adaptive cyber launcher icons
├── assets/
│   └── images/                # High-resolution app icon & branding assets
├── lib/
│   ├── app.dart              # Root MaterialApp & theme configuration
│   ├── main.dart              # App initialization (Firebase, Biometrics, Logging)
│   ├── core/                  # Core infrastructure layer
│   │   ├── config/            # Environment & backend URL configurations
│   │   ├── constants/         # AppColors, AppStrings, AppAssets
│   │   ├── network/           # ApiClient (JWT auto-auth), WebSocketService (with fallback)
│   │   ├── router/            # GoRouter with bottom navigation shell
│   │   ├── services/          # BiometricService, NotificationService, ReportPdfService
│   │   ├── storage/           # SecureStorageService (JWT token storage)
│   │   ├── theme/             # Cyber Dark theme tokens & Google Fonts
│   │   └── widgets/           # SGCard, SGButton, SGAppBar, SGStatisticCard, etc.
│   ├── features/              # Feature-First modular architecture
│   │   ├── ai/                # SecurePulse AI Agent chat screen & repository
│   │   ├── alerts/            # SOC incident alerts & incident detail investigation
│   │   ├── auth/              # Login screen with GitHub, Email, Biometric auth
│   │   ├── dashboard/         # Executive security dashboard & charts
│   │   ├── findings/          # Vulnerability findings & severity filters
│   │   ├── profile/           # Analyst profile & credentials
│   │   ├── reports/           # Executive compliance reports & PDF export
│   │   ├── repositories/      # Monitored codebases & SAST scan triggers
│   │   ├── scans/             # Scan history & detailed log viewer
│   │   ├── settings/          # Environment switching & platform settings
│   │   └── splash/            # Branded animated splash screen
│   └── providers/             # Riverpod providers & live stream notifiers
└── test/                      # Unit & integration test suite (15/15 passing)
```

## Complete Documentation Suite

Comprehensive technical, architectural, and operational documentation is available in the [docs/](#) directory and compiled to high-fidelity PDF dossiers:

| Document                                                                                                                                  | PDF Dossier                                                                                             | Description                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
|  <a href="#">Project Technical Report</a>                |  <a href="#">PDF</a>   | Master academic & technical report covering all 12 chapters, STRIDE threat model, and test results. |
|  <a href="#">Documentation Status &amp; Master Audit</a> |  <a href="#">PDF</a>   | Ground-truth engineering audit validating all 16 components against the actual codebase.            |
|  <a href="#">Evidence Index &amp; Register</a>           |  <a href="#">PDF</a>   | Master checklist indexing 48 evidence items (screenshots, code snippets, logs, videos, diagrams).   |
|  <a href="#">Technical Walkthrough &amp; Operations</a>  |  <a href="#">PDF</a>   | 30-section step-by-step engineering deployment and analyst operations manual.                       |
|  <a href="#">Product &amp; Engineering Roadmap</a>       |  <a href="#">PDF</a>   | Authoritative 17-phase development lifecycle and status delivery matrix.                            |
|  <a href="#">Master Architecture Mindmap</a>           |  <a href="#">PDF</a> | Visual 8-branch Mermaid mindmap, component hierarchy topology, and verification matrix.             |

## Getting Started

### Prerequisites

- [Flutter SDK](#) ≥ 3.0.0
- [Dart SDK](#) ≥ 3.0.0
- Android Studio / Android SDK Platform 34+

## Setup & Run

```
# 1. Clone repository
git clone https://github.com/NATTOMR/secureguard-mobile.git
cd secureguard-mobile

# 2. Fetch dependencies
flutter pub get

# 3. Run test suite
flutter test

# 4. Run on connected device or emulator
flutter run
```

## Building Release APK

```
flutter build apk --release
```

The compiled APK will be generated at `build/app/outputs/flutter-apk/app-release.apk` (and mirrored at `securepulse-release.apk` in the root folder).

## Backend Cloud Endpoints

The mobile app connects to the live **SecurePulse Cloud Service** hosted on Render:

| Endpoint                                | Method | Description                              |
|-----------------------------------------|--------|------------------------------------------|
| <code>/v1/auth/login</code>             | POST   | Analyst JWT authentication               |
| <code>/v1/dashboard</code>              | GET    | Overall security posture score & metrics |
| <code>/v1/alerts</code>                 | GET    | Active SOC security incidents            |
| <code>/ws/alerts</code>                 | WSS    | Real-time WebSocket threat stream        |
| <code>/v1/repositories</code>           | GET    | Monitored codebases telemetry            |
| <code>/v1/repositories/{id}/scan</code> | POST   | Trigger on-demand SAST security scan     |
| <code>/v1/ai/chat</code>                | POST   | Live AI Security Copilot query           |
| <code>/v1/reports</code>                | GET    | Executive compliance report logs         |

## Security & Privacy

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- **Zero-Trust Client Sessions:** Hardware-backed biometric authentication with auto-expiring JWT bearer tokens.
  - **Encrypted Transmission:** All cloud traffic enforced over **TLS 1.3 / WSS** with strict CORS and rate limiting.
  - **Push Protection:** Automated secret scanning to prevent credential leakage.
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## License

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